$$\mathbb{Z}/d\mathbb{Z} \times \mathbb{F}_{p^d} \longrightarrow \mathbb{F}_{p^d}$$
$$(i, \alpha) \longmapsto Fr^i(\alpha) = \alpha^{p^i}$$

$$\mathrm{Irr}_{|d}(\mathbb{F}_p) \xrightarrow{\sim} \left\{ \begin{matrix} \text{Orbits of} \\ \mathbb{Z}/d\mathbb{Z} \curvearrowright \mathbb{F}_{p^d} \end{matrix} \right\}$$

$$\begin{matrix} \cup\!\!\!\!\!\!\cdot\ \ \ \\ f(x) \longmapsto \left\{ \begin{matrix} \text{zeros of} \\ f(x) \text{ in } \mathbb{F}_{p^d} \end{matrix} \right\} \end{matrix}$$

$\deg r \mid d$

$\{\alpha, Fr(\alpha), Fr^2(\alpha), \dots, \}$

one zero

(2) says this is surjective

$$\begin{tikzcd}
\mathbb{F}_{p^d} \arrow[d] \arrow[dr, "\alpha \mapsto m_\alpha(x)"] & \\
\mathbb{F}_{p^d}/(\mathbb{Z}/d\mathbb{Z}) \arrow[r] & \mathrm{Irr}_{|d}(\mathbb{F}_p)
\end{tikzcd}$$

$$\mathbb{F}_{p^d}/(\mathbb{Z}/d\mathbb{Z}) = \left\{ \begin{matrix} \text{Orbits of} \\ \mathbb{Z}/d\mathbb{Z} \curvearrowright \mathbb{F}_{p^d} \end{matrix} \right\}$$

(4) says this is injective

Notation: $k \leq L$: field extension

$$\Updownarrow$$

$$L/k$$

NOT A QUOTIENT!

$$\underbrace{\mathrm{Aut}(L/k)} = \left\{ \begin{array}{l} \varphi : L \xrightarrow{\sim} L \text{ field auto.} \\ \text{s.t. } \varphi(a) = a,\ \forall a \in k \end{array} \right\}$$

Fact:

$$\mathbb{F}_{p^r} \leq \mathbb{F}_{p^d} \iff r \mid d$$

($\Longleftarrow$) $k = \mathbb{F}_{p^d}$: $\quad k_r = \{ \alpha \in \mathbb{F}_{p^d} : \underbrace{Fr^r(\alpha) = \alpha} \}$

then $|k_r| = p^r$.

$$\begin{tikzcd}
\mathbb{F}_{p^{18}} \arrow[d, no head] \arrow[drr, no head] & & \\
\mathbb{F}_{p^9} \arrow[dr, no head] & & \mathbb{F}_{p^6} \arrow[dl, no head] \arrow[dr, no head] & \\
& \mathbb{F}_{p^3} & & \mathbb{F}_{p^2}
\end{tikzcd}$$

$\mathbb{F}_p$

Linear transformations

Def^n k: field

V, W: k-vector spaces

A k-linear transformation

$$T : V \longrightarrow W$$

is a homomorphism of abelian groups s.t. $\forall a \in k,\ v \in V$

$$T(a \cdot v) = a \cdot T(v)$$

T is an isomorphism of k-vector spaces if it is bijective

Alt: T is invertible

Rmk: If T is an isomorphism, then $T^{-1}: W \to V$ is also an isomorphism of k-vector spaces

Pf: Need to check that T^{-1} is a k-linear transformation

- $\forall w_1, w_2 \in W$

$$T^{-1}(w_1 + w_2) \overset{?}{=} T^{-1}(w_1) + T^{-1}(w_2)$$

$\Leftarrow T(\cdot)$

$$w_1 + w_2 \overset{?}{=} T(T^{-1}(w_1) + T^{-1}(w_2))$$

T is a linear transformation $\to$ $\boxed{=}$ $T(T^{-1}(w_1)) + T(T^{-1}(w_2))$

$$\overset{\checkmark}{=} w_1 + w_2$$

- $\forall a \in k,\ w \in W$

$$T^{-1}(a\cdot w) \overset{?}{=} a\cdot T^{-1}(w)$$

$$\Downarrow T(\cdot)$$

$$a\cdot w \overset{?}{=} T(a\cdot T^{-1}(w))$$

$$\boxed{=}\ a\cdot T(T^{-1}(w))$$

$$\overset{\checkmark}{=} a\cdot w$$

Example:

1. $T: V \longrightarrow W$: zero or trivial

   $v \longmapsto 0$

2. $\mathrm{id}_V: V \longrightarrow V$: identity

   $v \longmapsto v$

3. $D: k[x] \longrightarrow k[x]$

   $f(x) \longmapsto f'(x)$

   $\sum a_i x^i \longmapsto \sum i a_i x^{i-1}$

4 $T : k^n \longrightarrow k^m$

Within k^n, we have

the standard basis vectors

$$e_1 = (1, 0, \dots 0)$$
$$e_2 = (0, 1, \dots 0)$$
$$\vdots$$
$$e_n = (0, \dots 0, 1)$$

Moreover, every vector $\underset{\in k^n}{\overset{v}{\overset{\shortparallel}{(a_1, a_2, \dots, a_n)}}}$

is equal to a linear combination

$$\begin{array}{ccc} v & = & a_1 e_1 + a_2 e_2 + \dots + a_n e_n \\ T \big\downarrow & & \big\downarrow T \\ T(v) & = & a_1 T(e_1) + a_2 T(e_2) + \dots + a_n T(e_n) \end{array}$$

Therefore, T is determined entirely by the vectors

$$T(e_1), T(e_2), \ldots, T(e_n)$$

Observation

(Homomorphism property of k^n)

Giving a k-linear transformation

$$T : k^n \longrightarrow W \leftarrow \text{any } k\text{-vector space}$$

is equivalent to giving n vectors

$$\begin{matrix} T(e_1), & T(e_2), & \ldots, & T(e_n) \in W \\ \| & \| & & \| \\ w_1 & w_2 & \ldots, & w_n \end{matrix}$$

T is given by

$$(a_1, a_2, .., a_n) \longmapsto a_1 w_1 + a_2 w_2 + \cdots + a_n w_n \in W$$

Defn. + Rmk. $T : V \to W$ k-linear transformation

$$\begin{tikzcd}
\underbrace{\ker T}_{k\text{-subspace}} \le V \arrow[d] \arrow[drr, "T"] & & \\
\underbrace{V/\ker T}_{k\text{-vector space}} \arrow[rr, "\text{isomorphism of } k\text{-vector spaces}"'] & & \underbrace{\operatorname{im} T \le W}_{k\text{-subspace}}
\end{tikzcd}$$

Rmk (1) $W \le V$: k-subspace

$$\begin{tikzcd}
V \arrow[d, "\pi"'] \\
V/W
\end{tikzcd}$$

is a k-linear transformation

Pf: Need to check

$$a \cdot (v + W) = a \cdot v + W$$

is well-defined

$w \in W \Rightarrow a \cdot w \in W$

$$\begin{array}{ccc} a \cdot (v + w + W) & = & a \cdot v + a \cdot w + W \\ \| & & \|\,?\,\checkmark \\ a \cdot (v + W) & = & a \cdot v + W \end{array}$$

Observation

$$T : k^n \longrightarrow W$$

$$\begin{array}{c} W \\ \cup \\ w_1, w_2, \dots, w_n \end{array}$$

$$T((a_1, \dots, a_n)) = \underbrace{a_1 w_1 + a_2 w_2 + \dots + a_n w_n}_{k\text{-linear combinations}}$$

Then

$$\operatorname{im} T = \langle \{w_1, w_2, \dots, w_n\} \rangle_k$$

$$\leq W$$

Def^n W is a finitely generated k-vector space if the following equiv. conditions hold:

(1) $\exists$ finite set $X \subseteq W$ s.t. $\langle X \rangle_k = W$

(2) $\exists$ a <u>surjective</u> k-linear transformation

$$T : k^n \longrightarrow W$$

for some $n \geq 1$.